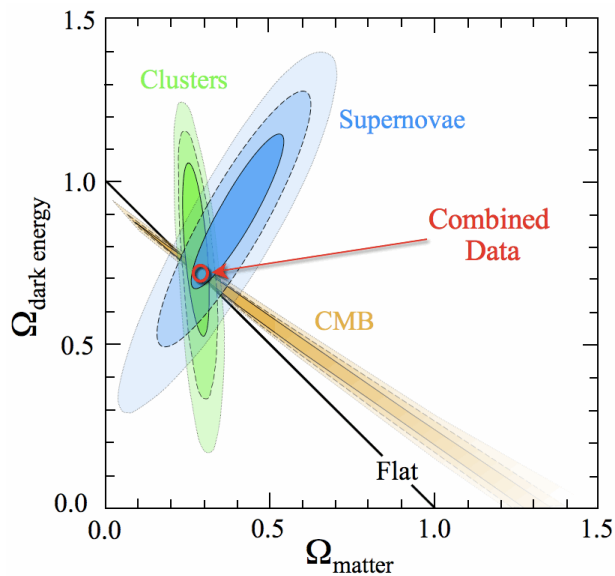
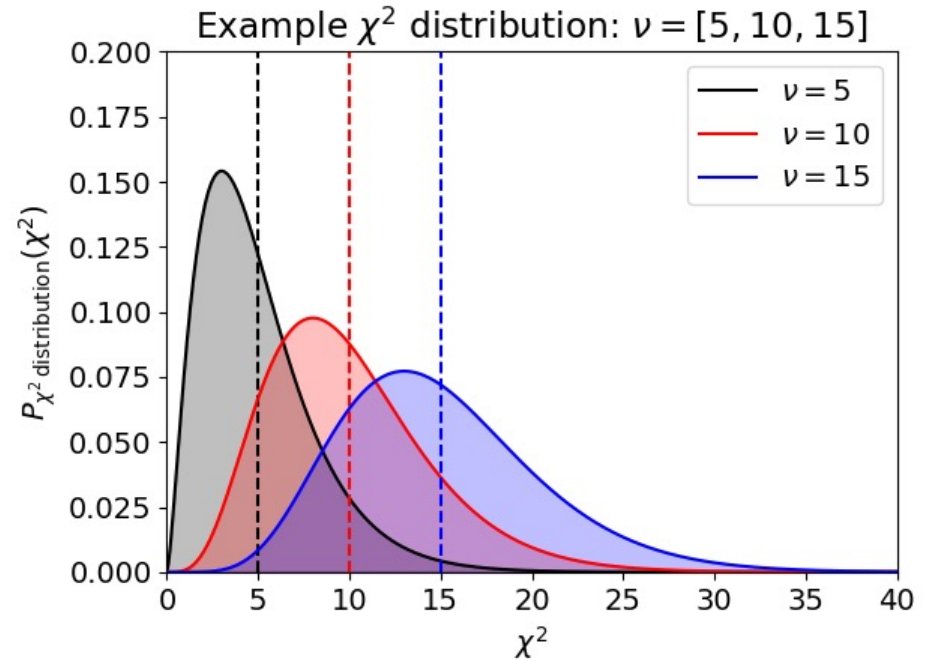
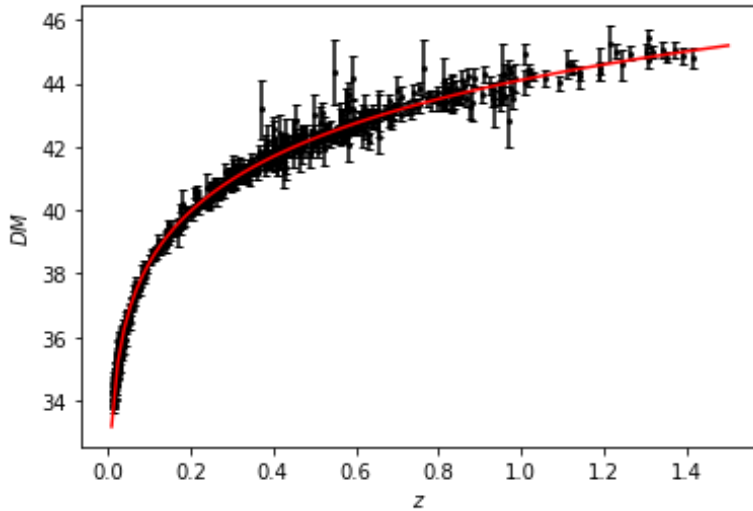
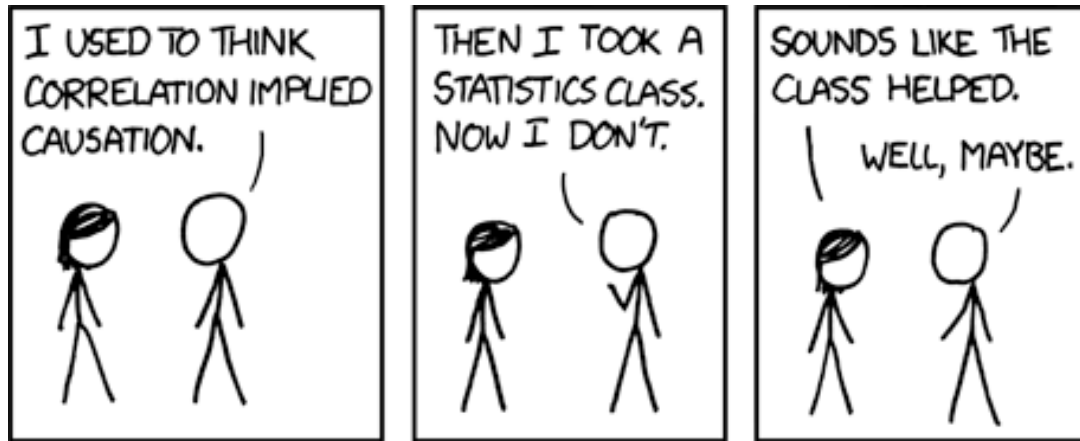


Class 3: Model-fitting

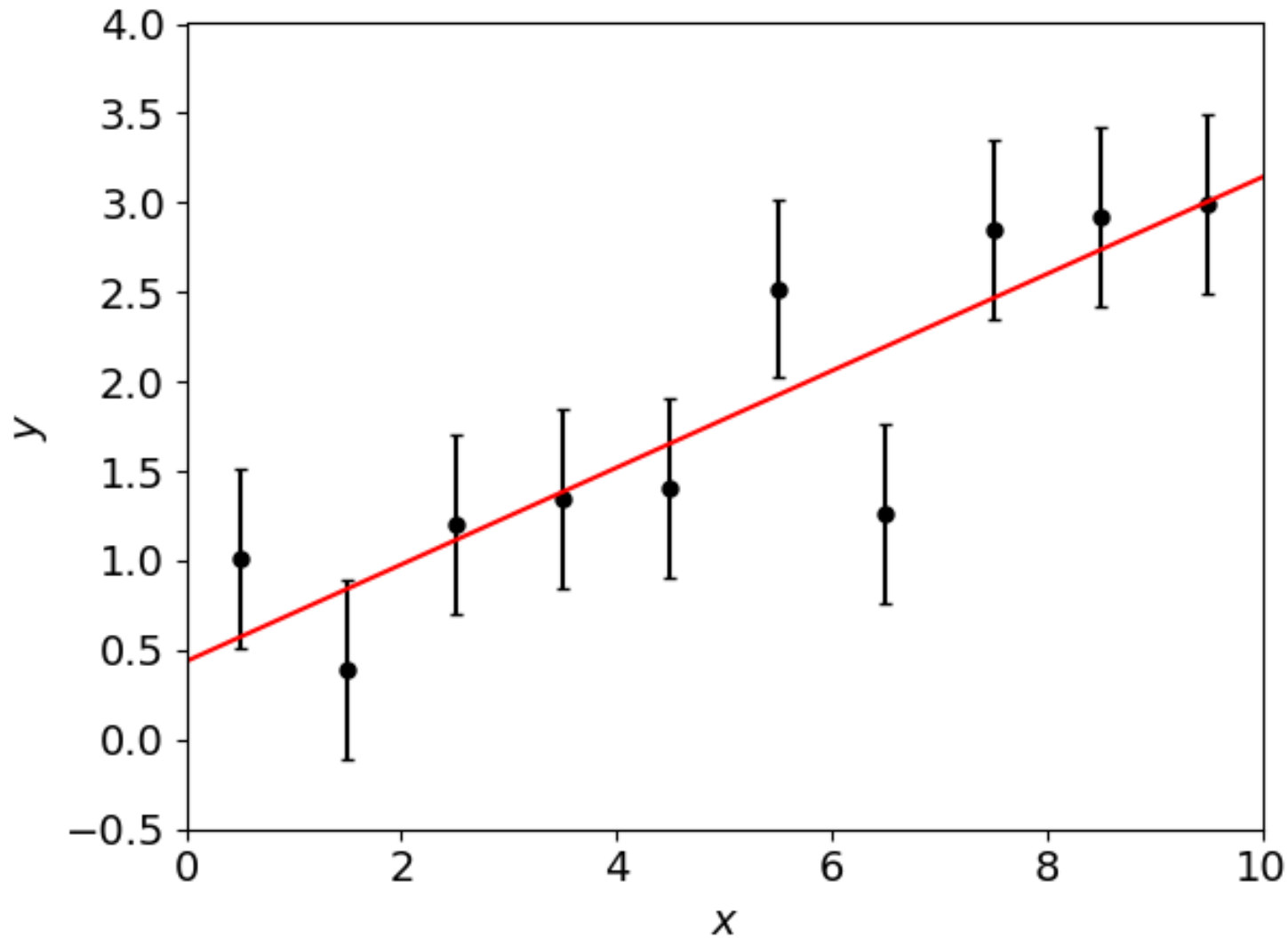


Class 3: Model-fitting



- Week 1: Probability and statistics
- Week 2: Correlation testing
- Week 3: Model fitting
- Week 4: Regression
- Week 5: Error estimates
- Week 6: Bayesian methods

Comparing data and models



Comparing data and models

- *When comparing data and models, we are typically doing one of two things ...*
- **Hypothesis testing:** we have a set of N measurements $x_i \pm \sigma_i$, which a theorist says should have values μ_i . *How probable is it that these measurements would have been obtained, if the theory is correct?*
- **Parameter estimation:** we have a parameterized model which describes the data, such as $y = ax + b$. *What are the best-fitting parameters and errors in those parameters?*

The χ^2 statistic

- The most important statistic to help with these tasks is the χ^2 **statistic** between the data $x_i \pm \sigma_i$ and model μ_i :

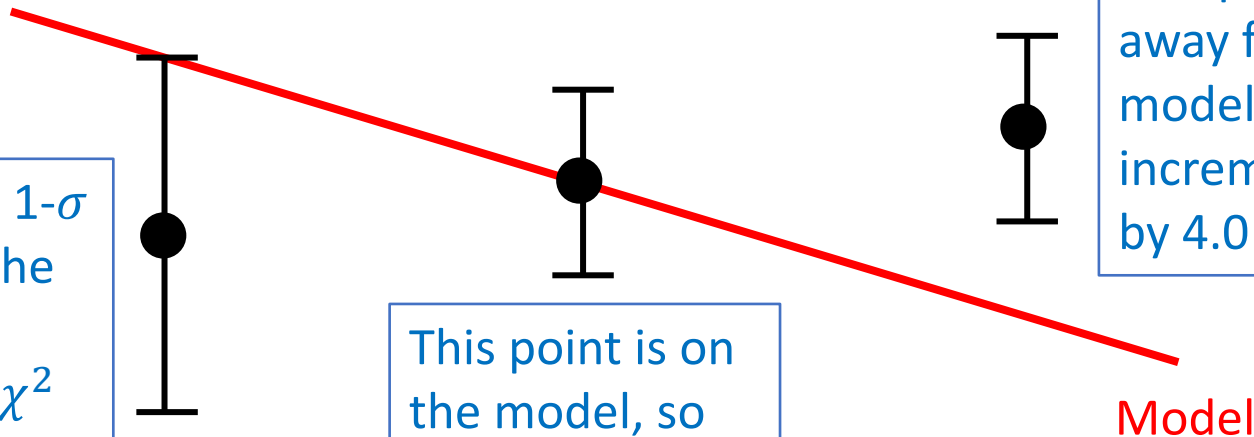
$$\chi^2 = \sum_{i=1}^N \left(\frac{x_i - \mu_i}{\sigma_i} \right)^2$$

χ^2 is a sum over the data points

This point is 1- σ away from the model, so increments χ^2 by 1.0

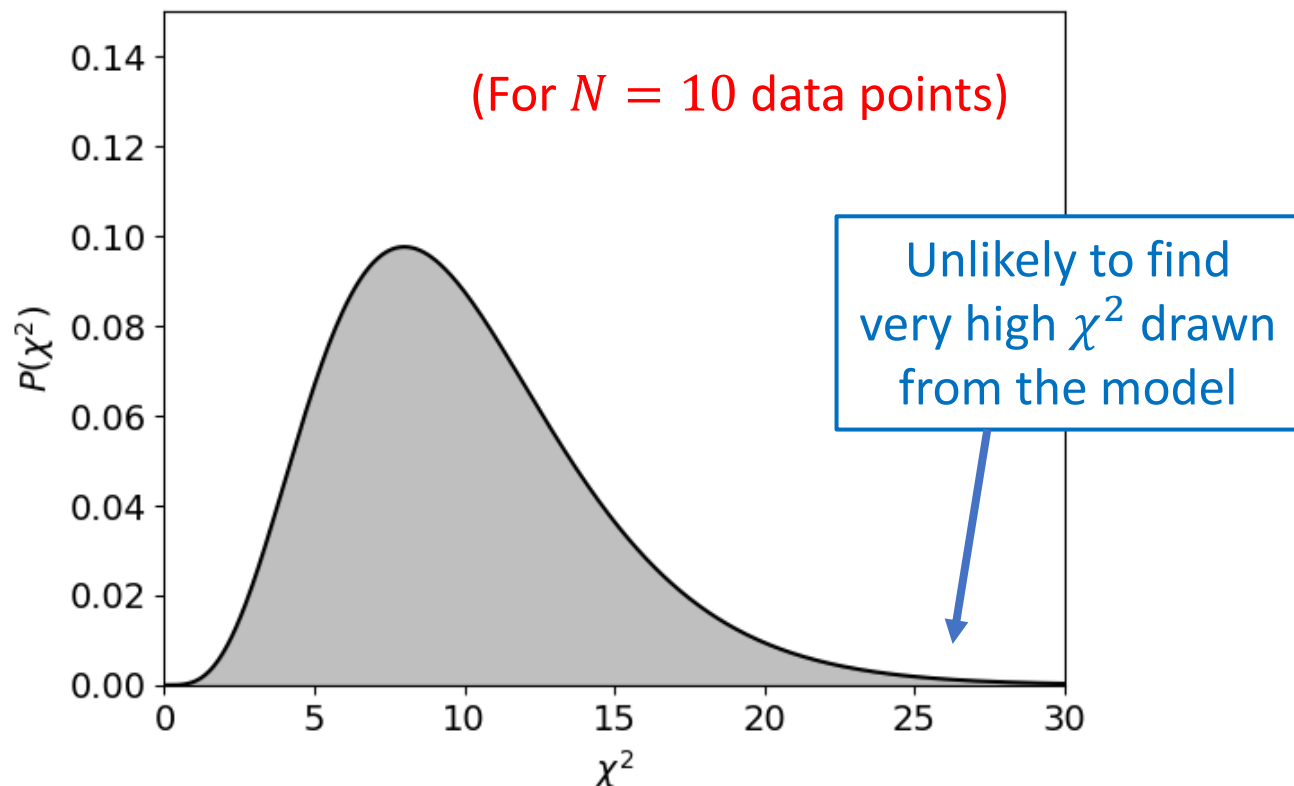
This point is on the model, so increments χ^2 by 0.0

This point is 2- σ away from the model, so increments χ^2 by 4.0



χ^2 probability distribution

- Sampling many realizations of N data points from a particular model, using Gaussian statistics, the χ^2 statistic has a **probability distribution**:



χ^2 probability distribution

- Sampling many realizations of N data points from a particular model, using Gaussian statistics, the χ^2 statistic has a **probability distribution**

$$P(\chi^2) \propto (\chi^2)^{\frac{\nu}{2}-1} e^{-\chi^2/2}$$

- ν is the number of **degrees of freedom** (number of data points minus number of fitted parameters)
- The **mean** of the distribution: $\overline{\chi^2} = \nu$
- The **variance**: $\text{Var}(\chi^2) = 2\nu$
- If the model is correct, we expect $\chi^2 \sim \nu \pm \sqrt{2\nu}$

χ^2 tests – examples

When testing a model, I find $\chi^2 = 2.3$ for $\nu = 20$ degrees of freedom. What action should I take?

When testing a model, I find $\chi^2 = 42.3$ for $\nu = 20$ degrees of freedom. What action should I take?

When testing a model, I find reduced $\chi^2/\nu = 1.5$. Can I reject the model?

I'm testing a model which predicts the number of objects in a series of bins, so I assign a Poisson error $\sqrt{N}$ for a χ^2 test. Why might this go wrong?

Use of χ^2 statistic as a hypothesis test

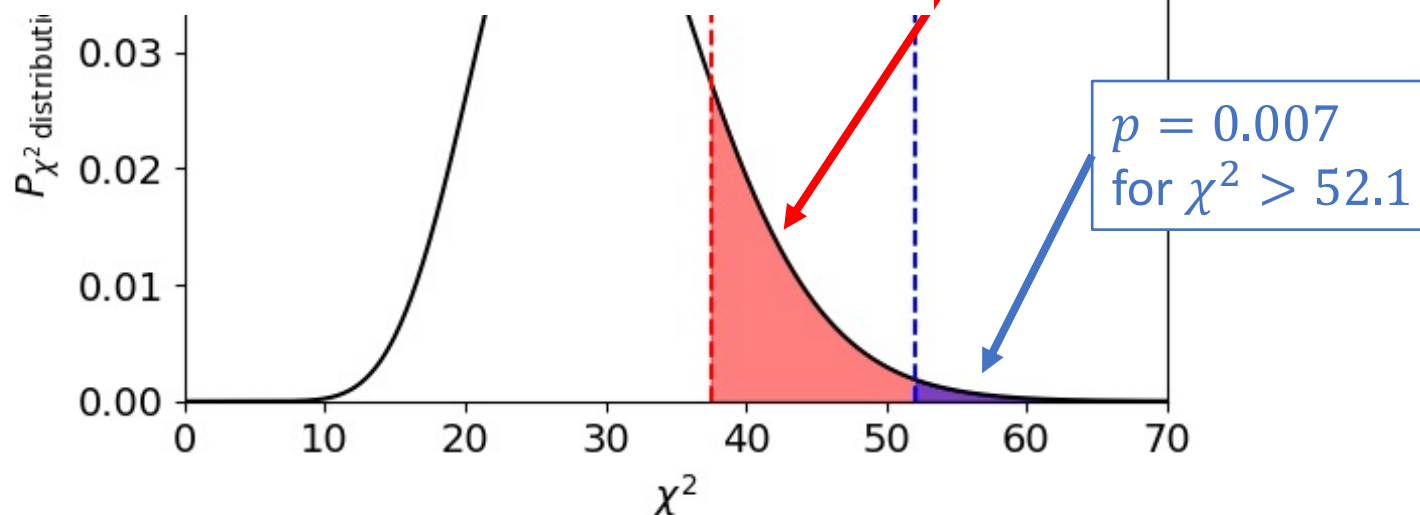
- *We can use the χ^2 statistic to construct a hypothesis test describing the “goodness of fit” between data and model*
- **Null hypothesis:** the data are consistent with the model
- **Test statistic:** χ^2
- **Distribution of values:** The χ^2 probability distribution
- **Confidence statement:** What is the probability that this value of χ^2 , or a larger one, could arise by chance?
- If the ***p*-value is not low**, the data are consistent with the model, which is *“ruled in”*
- If the ***p*-value is low**, the model is *“ruled out”*

Worked example: χ^2 as a hypothesis test

- Suppose that $\nu = 30$ and we have 2 datasets with $\chi^2 = 37.5$ and $\chi^2 = 52.1$. What are the corresponding p -values?

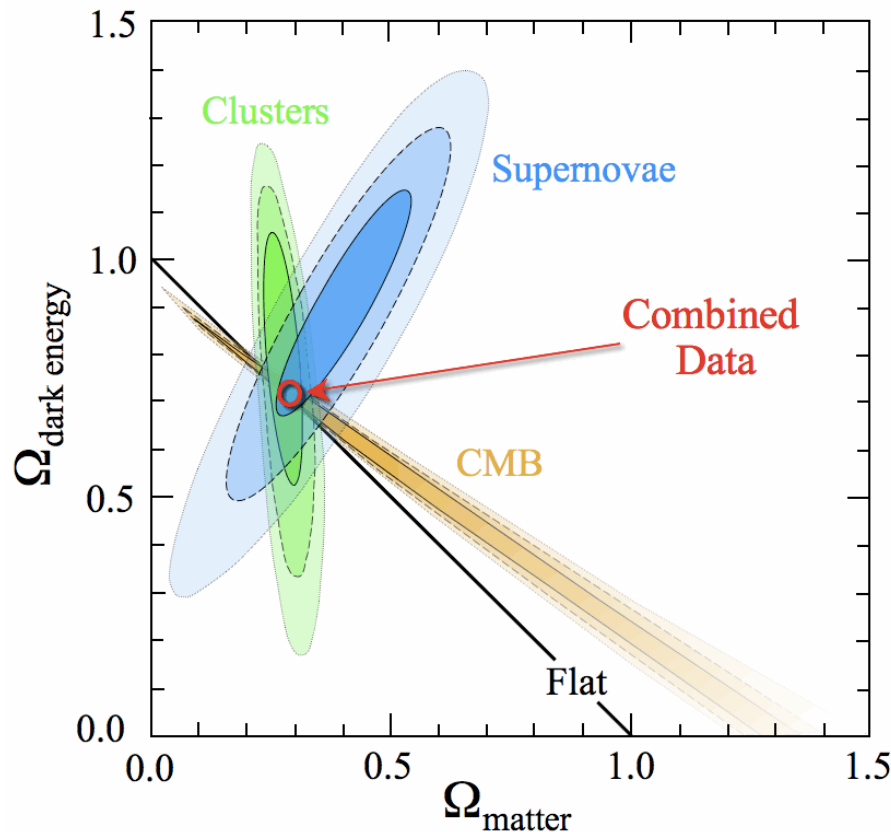
```
: nu = 30 # number of degrees of freedom
chisq1 = 37.5 # 1st value of chi-squared to test
chisq2 = 52.1 # 2nd value of chi-squared to test
p1 = stats.chi2.sf(chisq1,nu) # probability of chi-squared exceeding
p2 = stats.chi2.sf(chisq2,nu) # probability of chi-squared exceeding
print('p-value for chi2 =',chisq1,'and nu =',nu,'is p =',p1)
print('p-value for chi2 =',chisq2,'and nu =',nu,'is p =',p2)
```

```
p-value for chi2 = 37.5 and nu = 30 is p = 0.16301345461145947
p-value for chi2 = 52.1 and nu = 30 is p = 0.007429708671565914
```



Use of χ^2 statistic for parameter fitting

- A model typically contains free parameters. How do we determine the most likely values of these parameters and their error ranges?*

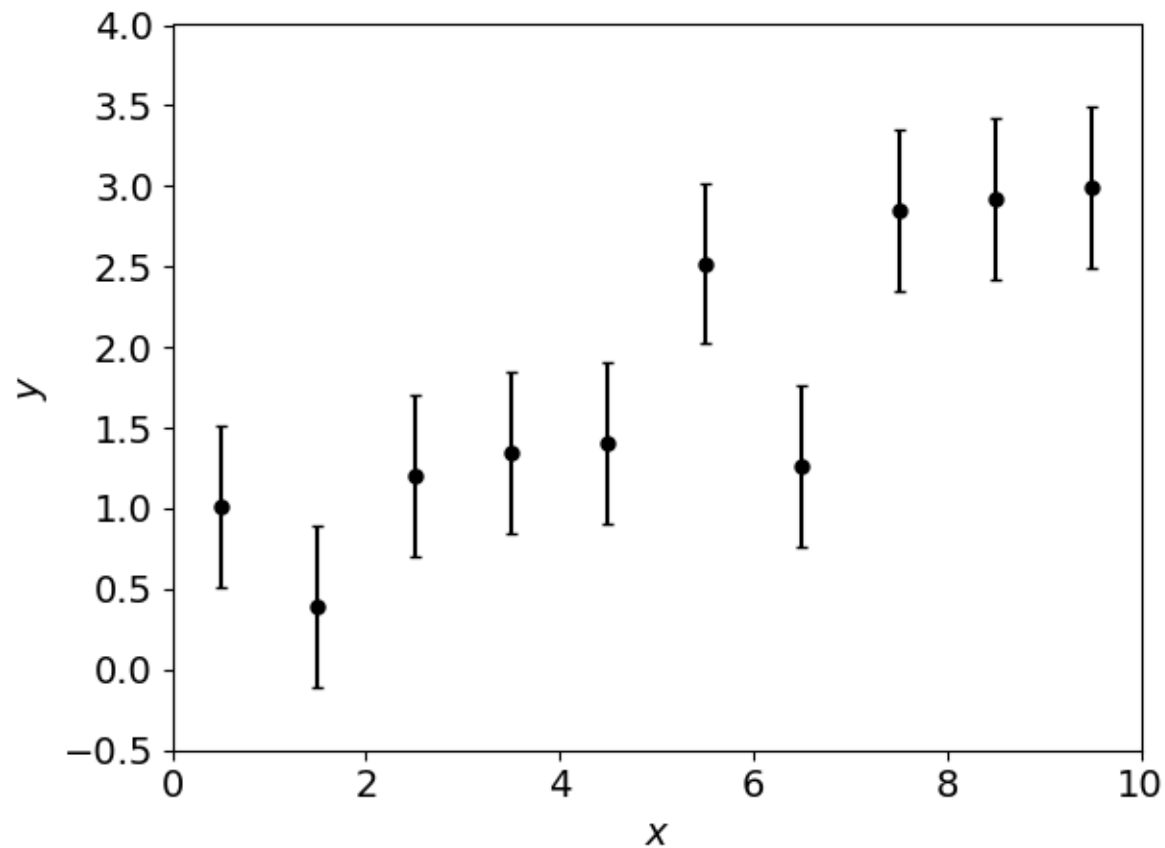


Use of χ^2 statistic for parameter fitting

- *A model typically contains free parameters. How do we determine the most likely values of these parameters and their error ranges?*
- Suppose we are fitting a model with 2 free parameters (a, b)
- The most likely (“best-fitting”) values of (a, b) are found by **minimizing the χ^2 statistic**
- The joint error distribution of (a, b) can be found by **calculating the values of χ^2 over a grid** of (a, b) and enclosing a particular region $\chi^2 < \chi^2_{\min} + \Delta\chi^2$

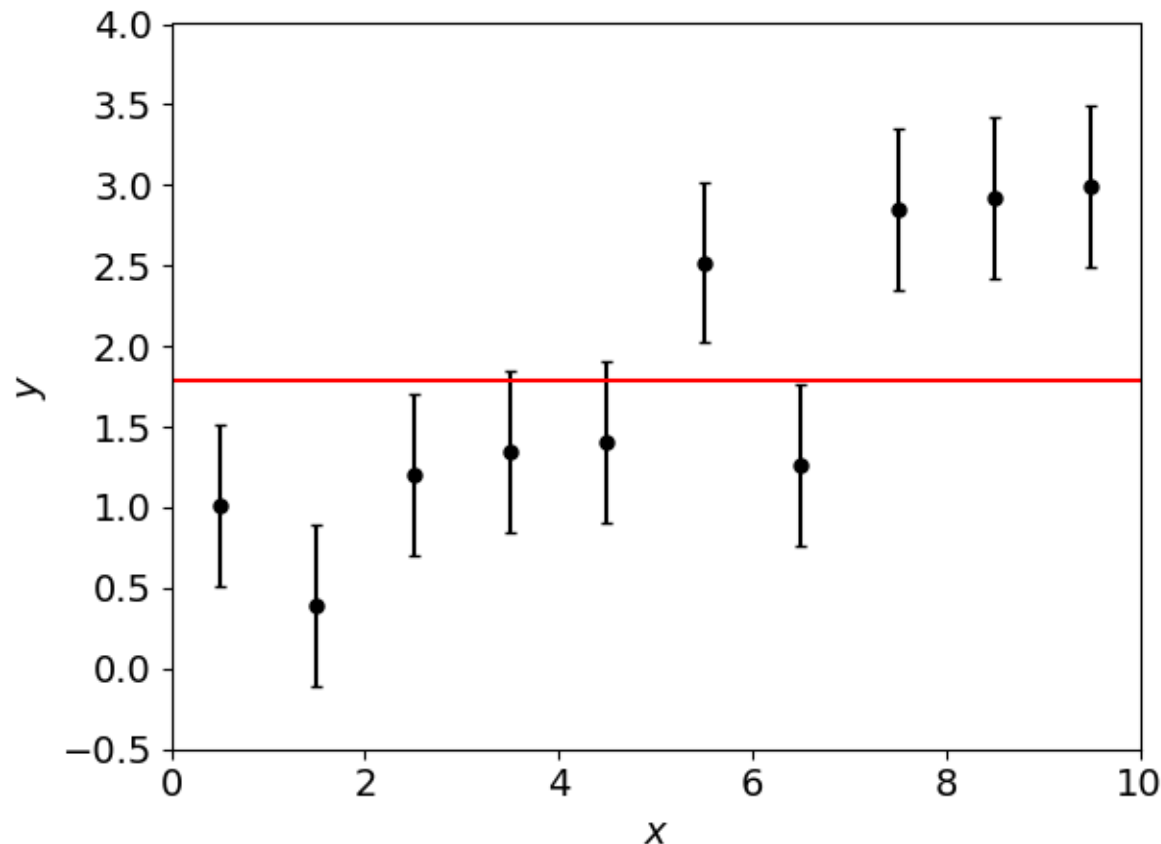
Worked example: χ^2 for parameter fitting

- Here is an example dataset containing $N = 10$ points:



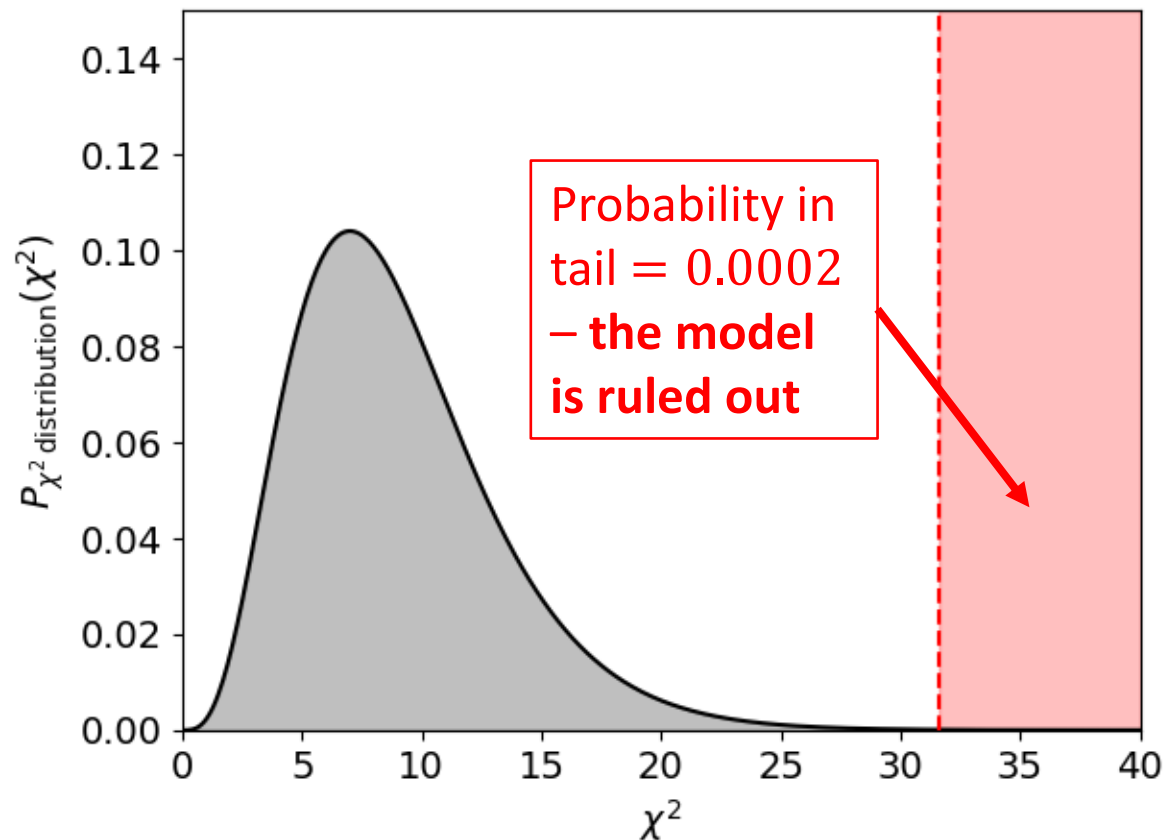
Worked example: χ^2 for parameter fitting

- *Could these points be fit by a constant $y = b$?*
- Minimizing χ^2 , we find $\chi^2_{\min} = 31.6$ for $b = 1.79$



Worked example: χ^2 for parameter fitting

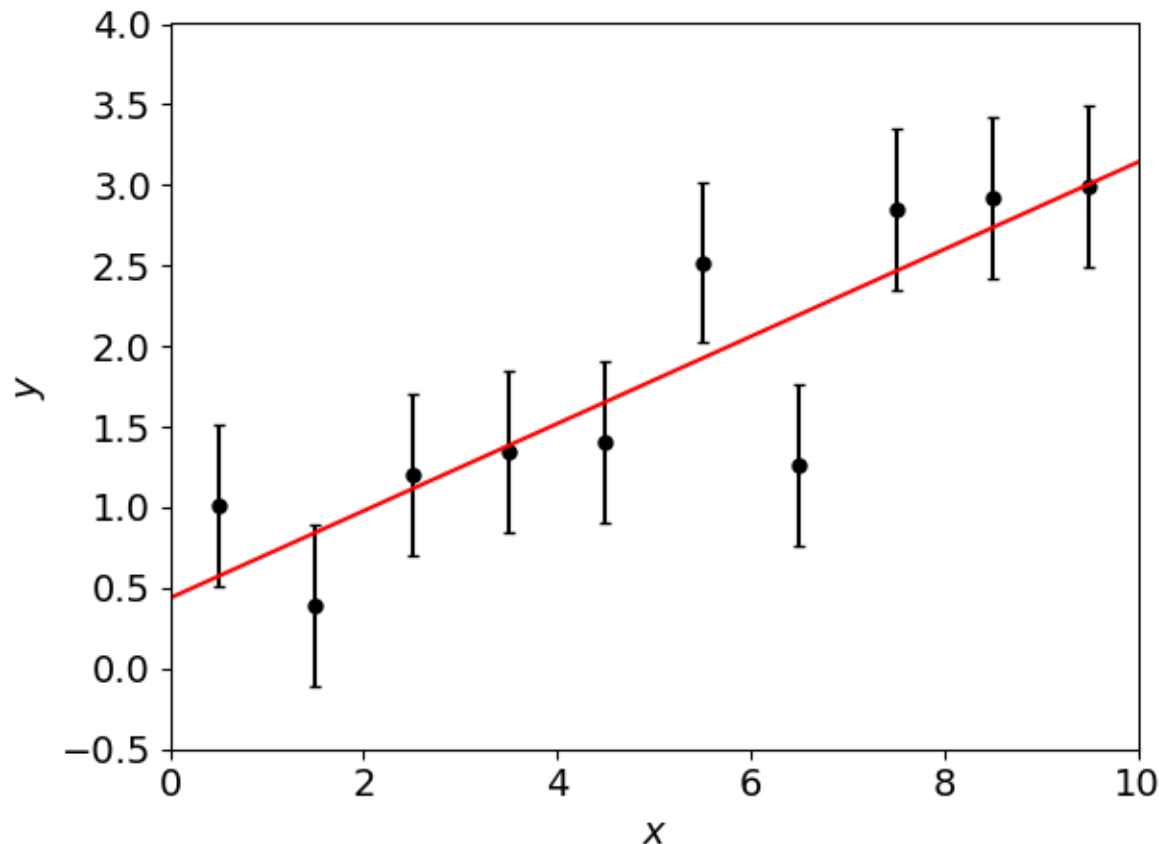
- *Is the minimum χ^2 likely given the model?* Consider the χ^2 probability distribution for $\chi^2 > 31.6$ and $\nu = N - 1 = 9$:



Worked example: χ^2 for parameter fitting

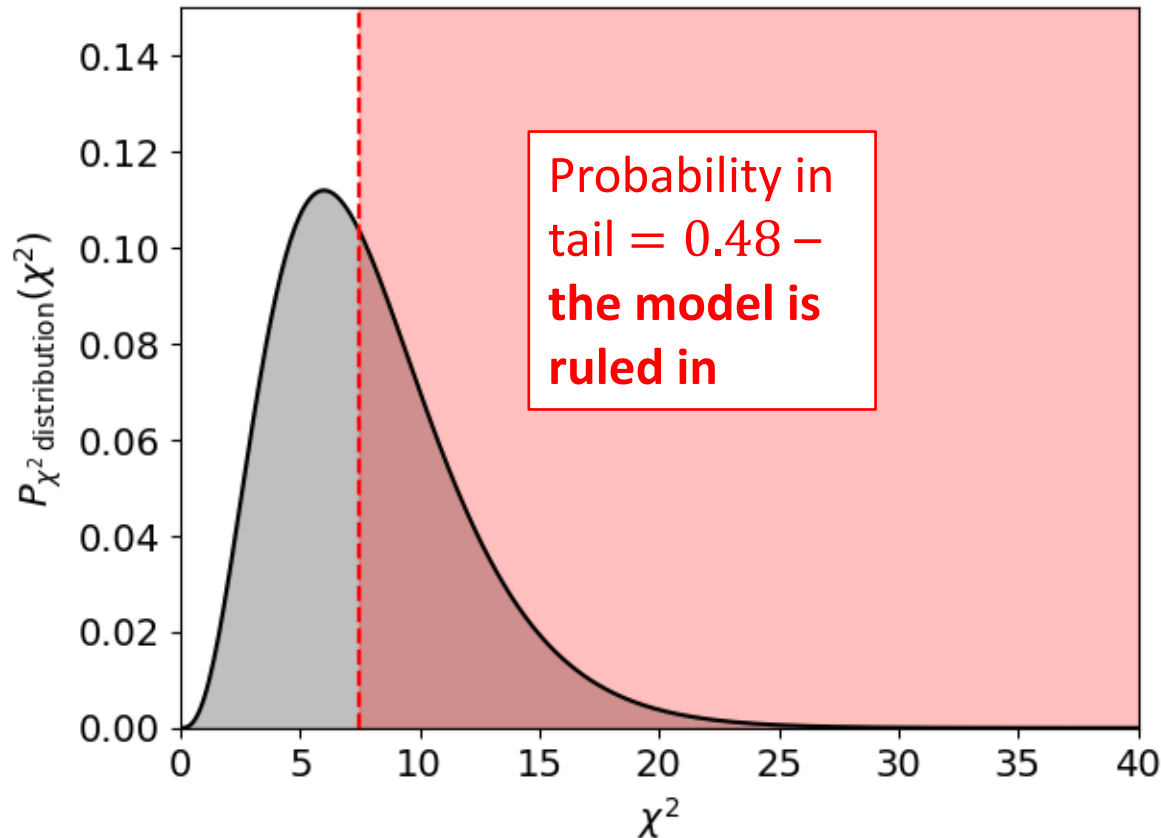
- Could these points be fit by a straight line $y = ax + b$?*

Minimizing χ^2 , we find $\chi^2_{\min} = 7.5$ for $a = 0.27$ and $b = 0.44$



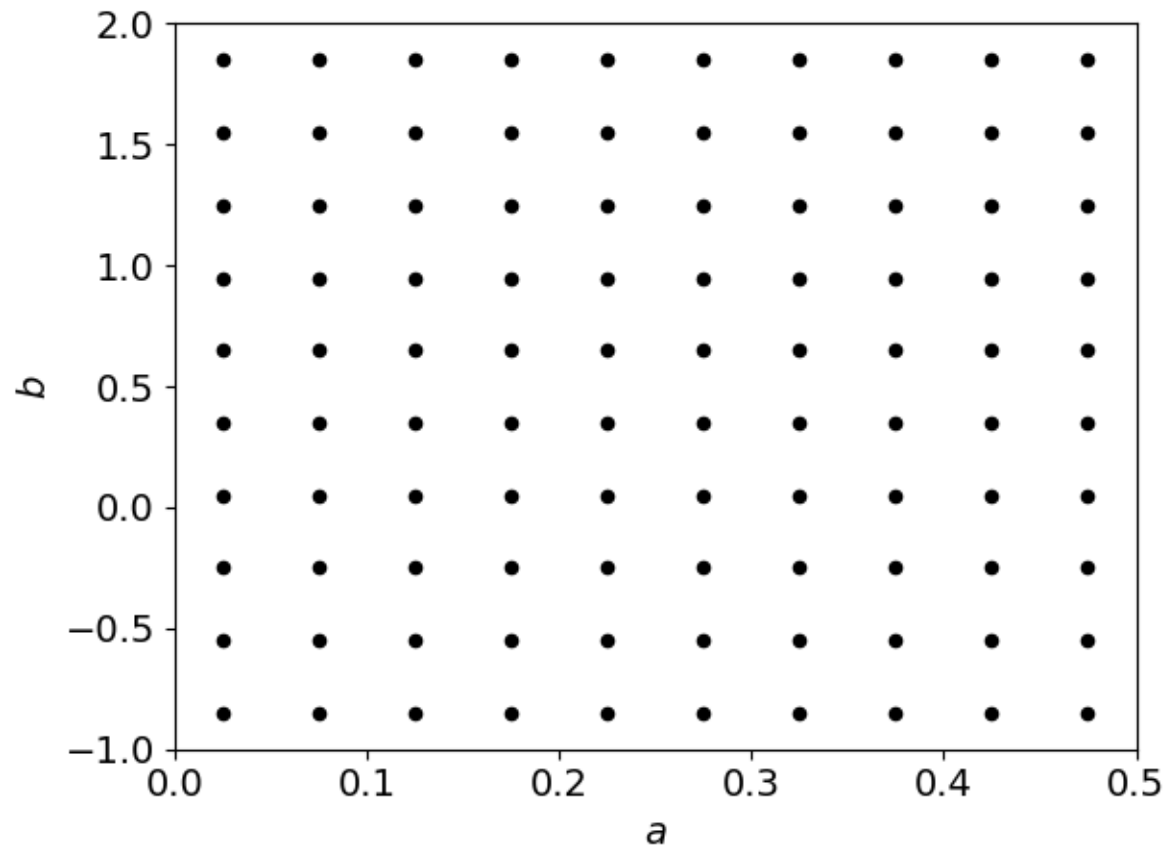
Worked example: χ^2 for parameter fitting

- *Is the minimum χ^2 likely given the model?* Consider the χ^2 probability distribution for $\chi^2 > 7.5$ and $\nu = N - 2 = 8$:



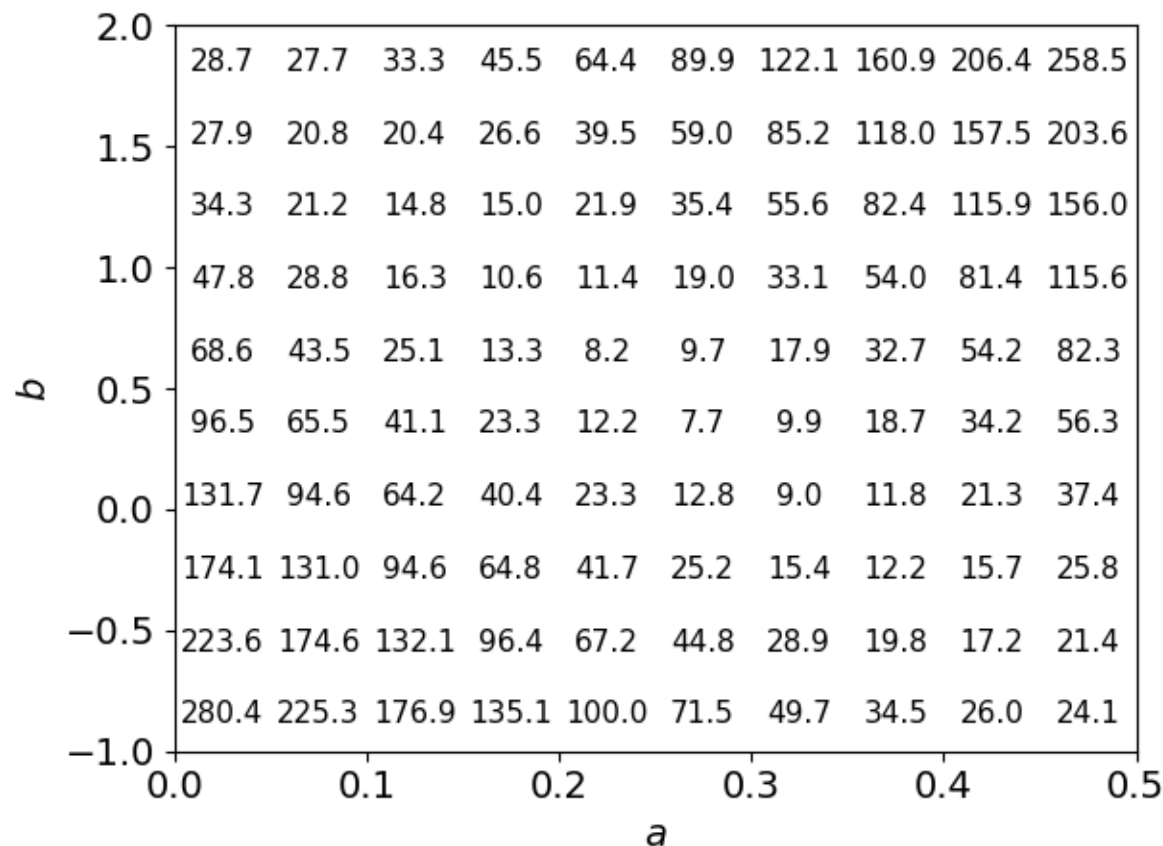
Worked example: χ^2 for parameter fitting

- *Now let's determine the error ranges.* What is the distribution of χ^2 values across the (a, b) grid?



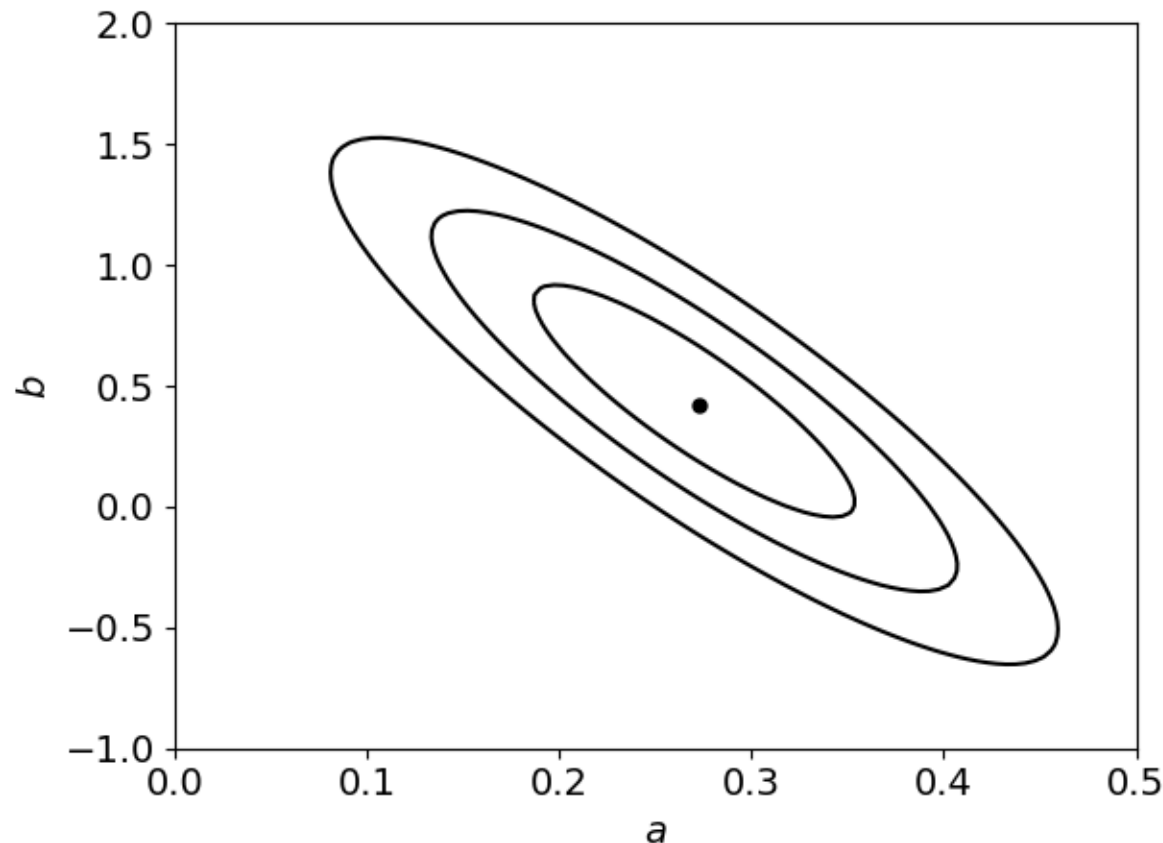
Worked example: χ^2 for parameter fitting

- *Now let's determine the error ranges.* What is the distribution of χ^2 values across the (a, b) grid?



Worked example: χ^2 for parameter fitting

- *Set confidence regions using $\Delta\chi^2$ intervals:* for 2 parameters, the (68,95,99)% regions are $\Delta\chi^2 = (2.30, 6.17, 11.8)$



Worked example: χ^2 for parameter fitting

- *Set confidence regions using $\Delta\chi^2$ intervals:* for 2 parameters, the (68,95,99)% regions are $\Delta\chi^2 = (2.30, 6.17, 11.8)$

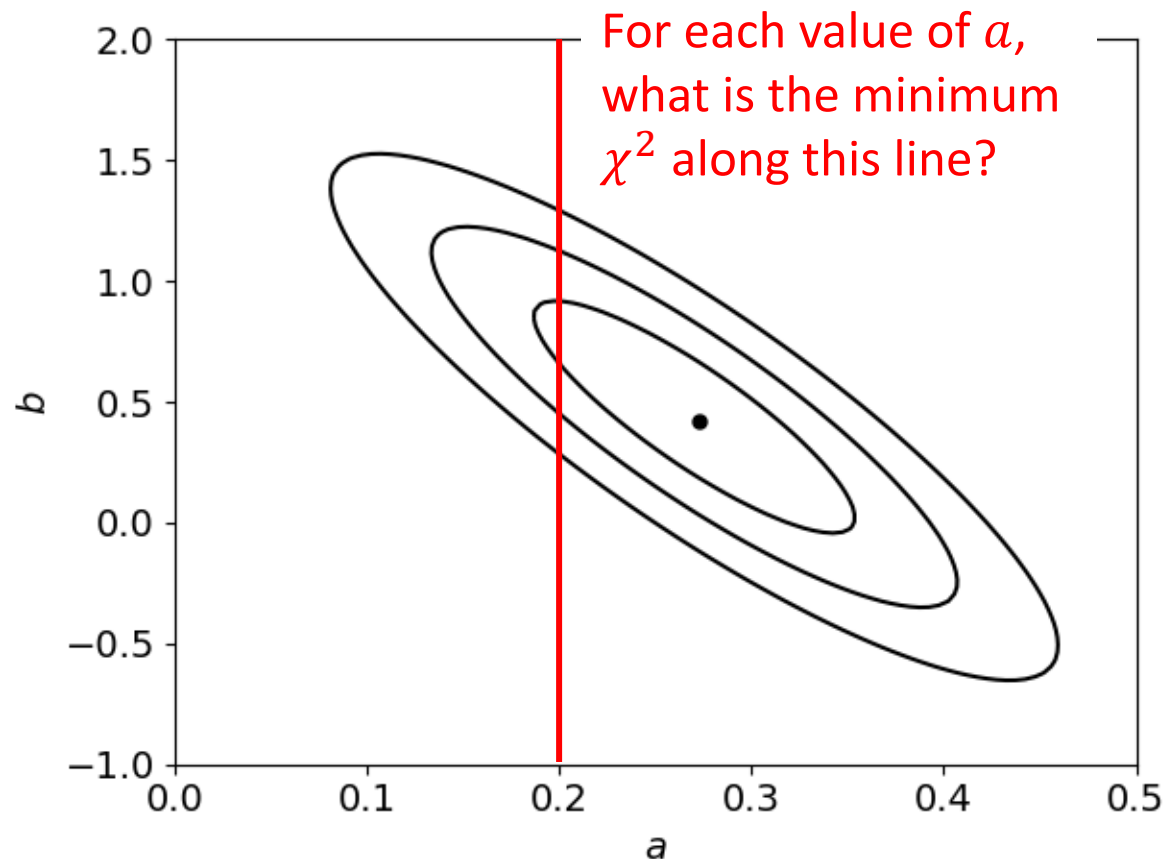
$\Delta\chi^2$ as a Function of Confidence Level and Degrees of Freedom						
	ν					
p	1	2	3	4	5	6
68.3%	1.00	2.30	3.53	4.72	5.89	7.04
90%	2.71	4.61	6.25	7.78	9.24	10.6
95.4%	4.00	6.17	8.02	9.70	11.3	12.8
99%	6.63	9.21	11.3	13.3	15.1	16.8
99.73%	9.00	11.8	14.2	16.3	18.2	20.1
99.99%	15.1	18.4	21.1	23.5	25.7	27.8

2 parameters varying

1- σ
confidence

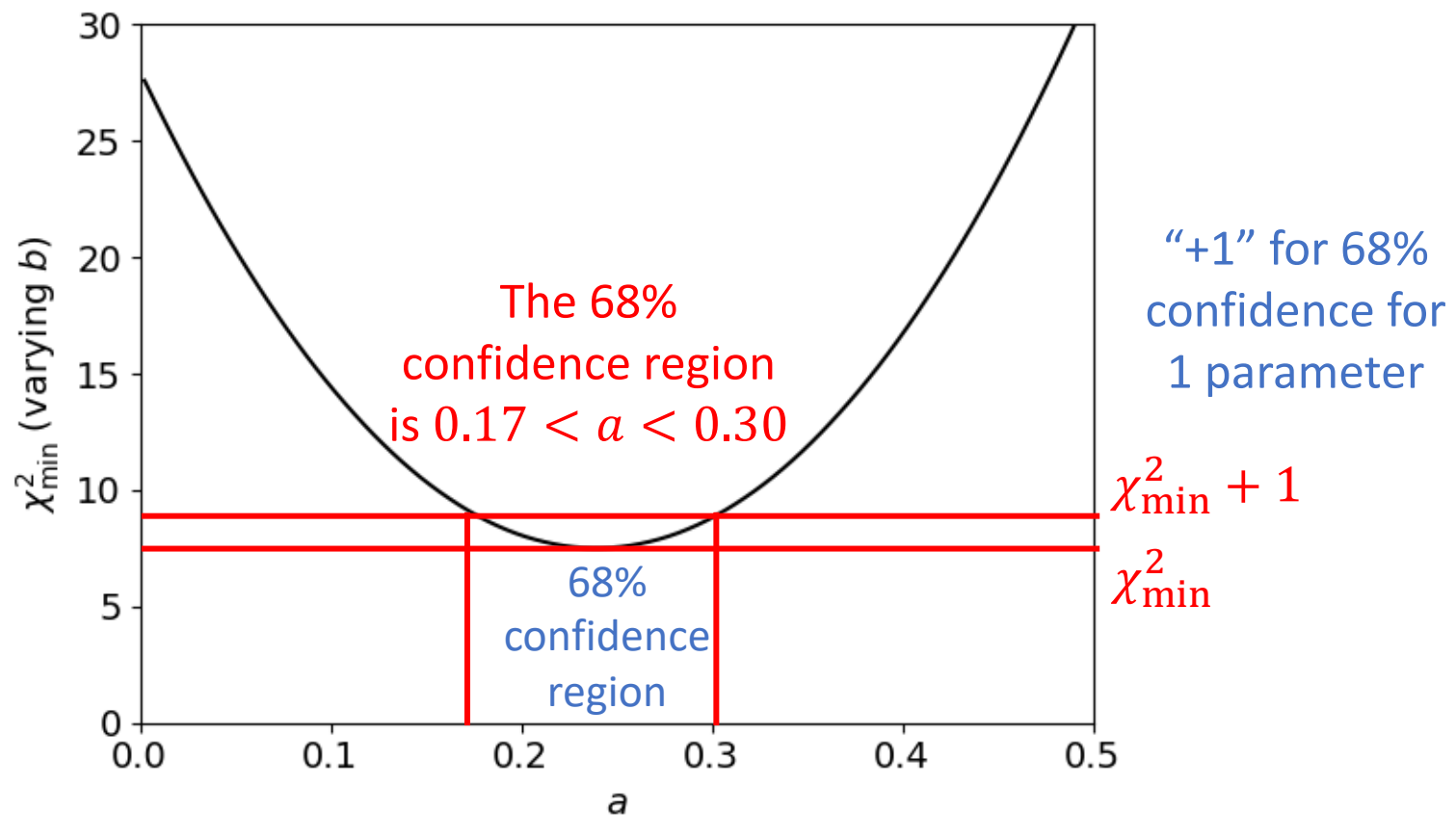
Worked example: χ^2 for parameter fitting

- *What about errors in individual parameters?* For each value of parameter a , find the minimum χ^2 varying parameter b :



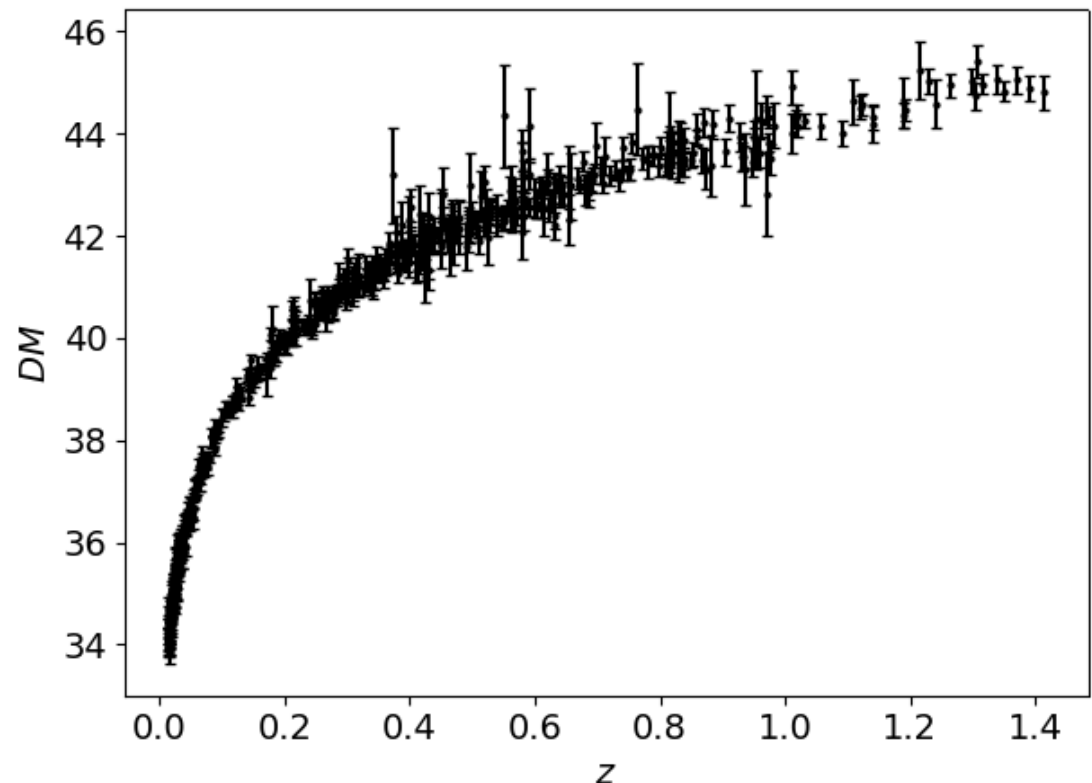
Worked example: χ^2 for parameter fitting

- What about errors in individual parameters?* For each value of parameter a , find the minimum χ^2 varying parameter b :



Activity: supernova cosmology

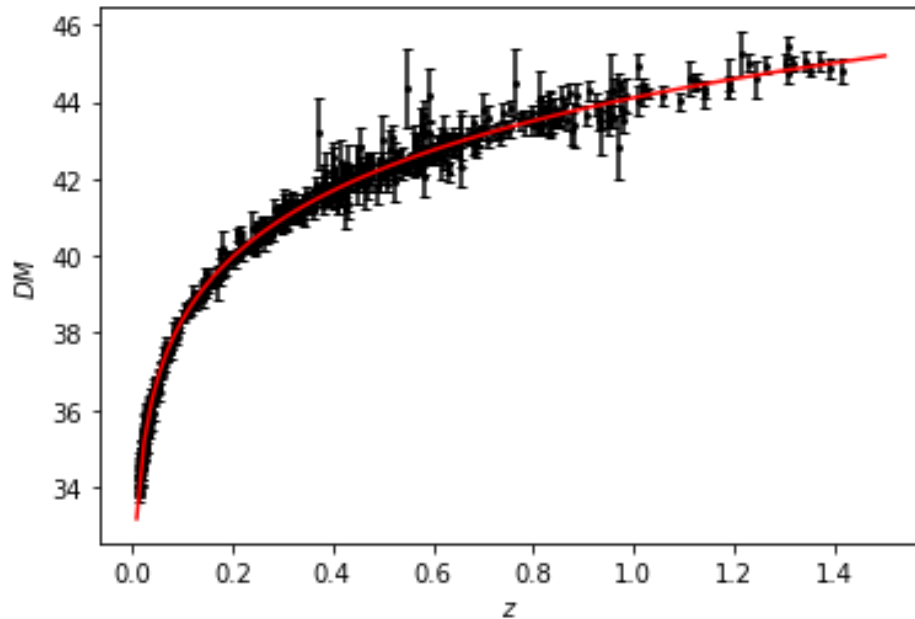
- In this Activity we will use a recent **supernova distance-redshift dataset** to determine the parameters $(\Omega_m, \Omega_\Lambda)$
- Minimize χ^2 and find the best-fitting $(\Omega_m, \Omega_\Lambda)$
- Construct the **2D confidence regions** in $(\Omega_m, \Omega_\Lambda)$
- Determine the **individual errors** in Ω_m and Ω_Λ



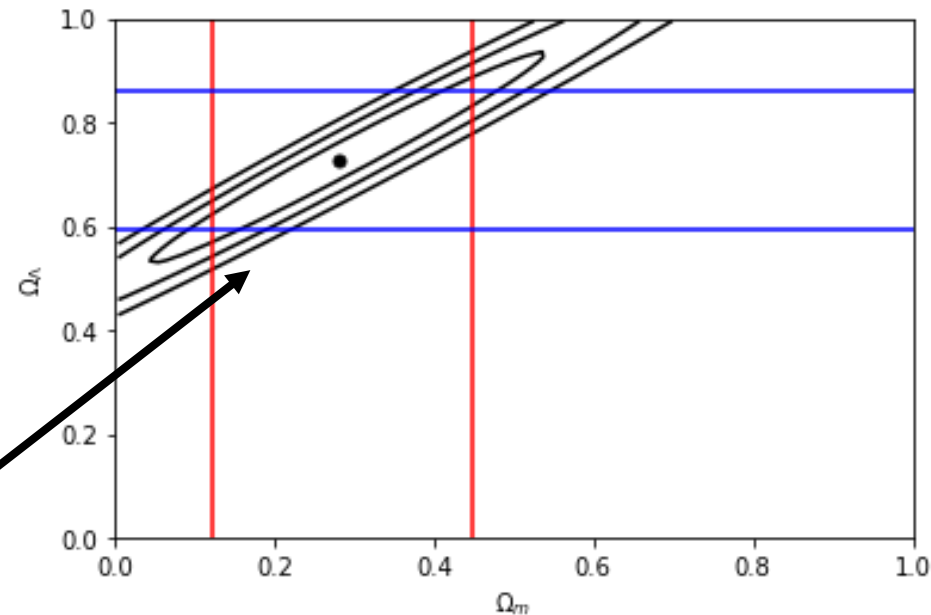
Activity: supernova cosmology

Best-fitting model:

$$\Omega_m = 0.28, \Omega_\Lambda = 0.73, \chi^2 = 562.2, \nu = 578$$



68% confidence for Ω_m : 0.12 – 0.45
68% confidence for Ω_Λ : 0.59 – 0.86



$\Delta\chi^2 = (2.30, 6.17, 11.8)$
for 2 parameters